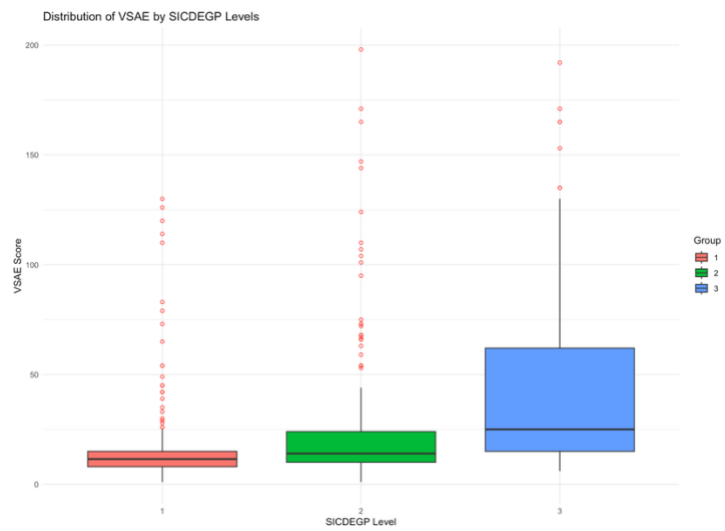
Project 1 – HW8

a) I used the book's code and grouped the data by sicdegp level, and printed a table of the vsae standard deviations. Remembering that 0 means group 3, we can see that group 3 has a significantly higher standard deviation than 1 and 2, being ordered from 1 – 2 – 0 in order of increasing variance. My boxplots are as follows, which certainly confirm this data from the table. One comment I can make is that in the first two groups, there are certainly many outliers (indicated by the red dots). The means also increase across levels, but the most significant difference is the width of the boxes representing standard deviation.

```
       0        1        2
37.53372 21.39158 28.64497
```



b) After refitting the model with the updated residual structure, we get this R summary. Based on the variance function summary, we can very clearly see this trend reflected in the variance adjustments by group. The variance from group 3 (lvl 0) to group 1 gets cut in half! Same with group 3 to group 2, although cut by a little less than a half (but not by much compared to group 1). Intuitively this certainly tells us that this updated structure will be better, and an anova LRT between the two models confirms this with a p-value of less than .0001 under a null distribution assumption of Chi^2 with df = 2. AIC and BIC also significantly support this decision, even though we make our model more complex, the fit is so much better that this difference is also reflected in the BIC measurement.

```
Linear mixed-effects model fit by REML
  Data: autism.grouped
       AIC      BIC    logLik
  4561.834 4619.059 -2267.917

Random effects:
 Formula: ~age.2 + I(age.2^2) - 1 | childid
 Structure: General positive-definite, Log-Cholesky parametrization
            StdDev    Corr
age.2      4.2589716 age.2
I(age.2^2) 0.3745326 -0.448
Residual   9.0995357

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | sicdegp2.f
 Parameter estimates:
        0         1         2
1.0000000 0.4910024 0.5089080
Fixed effects:  vsae ~ age.2 + I(age.2^2) + sicdegp2.f + age.2:sicdegp2.f
                      Value Std.Error  DF   t-value p-value
(Intercept)       13.228235 1.1146097 448 11.868042  0.0000
age.2              6.530380 0.7546568 448  8.653443  0.0000
I(age.2^2)         0.081189 0.0425100 448  1.909878  0.0568
sicdegp2.f1       -4.823575 1.2159177 155 -3.967024  0.0001
sicdegp2.f2       -3.470437 1.1940808 155 -2.906367  0.0042
age.2:sicdegp2.f1 -4.294852 0.9047568 448 -4.746969  0.0000
age.2:sicdegp2.f2 -3.701015 0.8575486 448 -4.315808  0.0000
 Correlation:
                  (Intr) age.2  I(.2^2 scd2.1 scd2.2 a.2:2.1
age.2             -0.365
I(age.2^2)         0.173 -0.415
sicdegp2.f1       -0.904  0.305 -0.087
sicdegp2.f2       -0.920  0.309 -0.085  0.837
age.2:sicdegp2.f1  0.257 -0.720  0.070 -0.293 -0.234
age.2:sicdegp2.f2  0.269 -0.756  0.065 -0.242 -0.302  0.612

Standardized Within-Group Residuals:
        Min         Q1        Med         Q3        Max
-2.8510354 -0.4077998 -0.0458251  0.3004446  5.8409745

Number of Observations: 610
Number of Groups: 158
```

```
> anova(model6.3.fit, model6.4.fit)
             Model df      AIC      BIC    logLik   Test  L.Ratio p-value
model6.3.fit     1 11 4633.570 4681.991 -2305.785
model6.4.fit     2 13 4561.834 4619.059 -2267.917 1 vs 2 75.73606  <.0001
```

```
> anova(Model.better.ml, Model.test.ml)
               Model df      AIC      BIC    logLik   Test  L.Ratio p-value
Model.better.ml     1 13 4562.231 4619.606 -2268.115
Model.test.ml       2 14 4534.921 4596.710 -2253.461 1 vs 2 29.30937  <.0001
```

c) I denoted this second model as Model.better and then updated it and named a new model Model.test, updating both models to now fit using ML since we are going to try adding a cubic fixed effect term. Based on an anova LRT following a Chi^2 with df = 1, a p-value of <.0001 tells us that the inclusion of this cubic term significantly improves our model fit. AIC and BIC both support this coice as well, again making the model more complex improves the fit so much that the BIC criterion still decreases even taking into consideration the added penalty for the extra term. Model.test with the cubic term is certainly the best way to go.

```
> summary(Model.best)
Linear mixed-effects model fit by REML
  Data: autism.grouped
      AIC      BIC    logLik
  4542.69 4604.294 -2257.345

Random effects:
 Formula: ~age.2 + I(age.2^2) - 1 | childid
 Structure: General positive-definite, Log-Cholesky parametrization
            StdDev    Corr
age.2      4.4063494 age.2
I(age.2^2) 0.3863845 -0.482
Residual   9.0518482

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | sicdegp2.f
 Parameter estimates:
         0         1         2
1.0000000 0.4477601 0.4897606
Fixed effects:  vsae ~ age.2 + I(age.2^2) + sicdegp2.f + I((age - 2)^3) + age.2:sicdegp2.f
                       Value Std.Error  DF   t-value p-value
(Intercept)       12.212929 1.1251892 447 10.854112  0.0000
age.2              9.069647 0.8814090 447 10.289941  0.0000
I(age.2^2)        -0.589063 0.1259635 447 -4.676456  0.0000
sicdegp2.f1       -4.693866 1.1960070 155 -3.924614  0.0001
sicdegp2.f2       -3.396562 1.1847979 155 -2.866786  0.0047
I((age - 2)^3)     0.042127 0.0074713 447  5.638505  0.0000
age.2:sicdegp2.f1 -4.389117 0.9084657 447 -4.831351  0.0000
age.2:sicdegp2.f2 -3.776408 0.8620045 447 -4.380961  0.0000
 Correlation:
                  (Intr) age.2  I(.2^2 scd2.1 scd2.2 I((-2) a.2:2.1
age.2             -0.384
I(age.2^2)         0.207 -0.599
sicdegp2.f1       -0.909  0.272 -0.051
sicdegp2.f2       -0.915  0.268 -0.041  0.854
I((age - 2)^3)    -0.160  0.506 -0.940  0.020  0.012
age.2:sicdegp2.f1  0.250 -0.628  0.039 -0.280 -0.230 -0.015
age.2:sicdegp2.f2  0.261 -0.657  0.035 -0.239 -0.291 -0.014  0.613

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-2.66483469 -0.40971559 -0.06558837  0.30801656  6.01766768

Number of Observations: 610
```

d) Given parts b and c, we certainly have a better model than the one in the book. After refitting using REML, here is the output summary. We can see that with this new added term, the difference in variance among the Sicdegp groups is even more distinct, reducing by 44% and 49% compared to 49% and 50% respectively among groups 1 and 2. All our fixed effect terms are highly significant in their contributions to the mean value, and our random effect variances and correlations do not change by a significant margin, telling us this fit is straight up better.

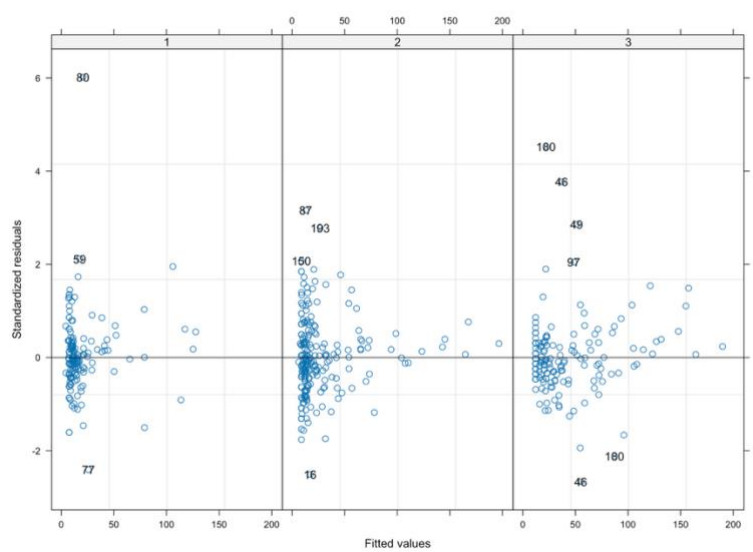
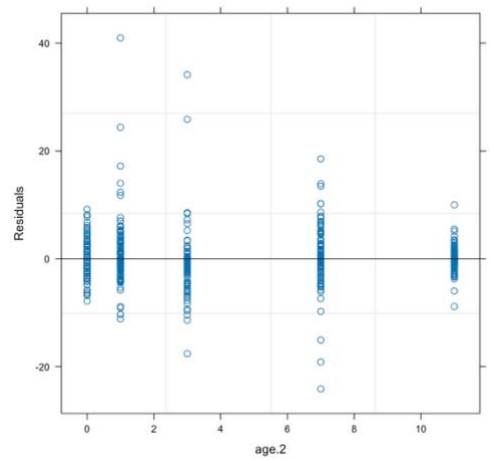


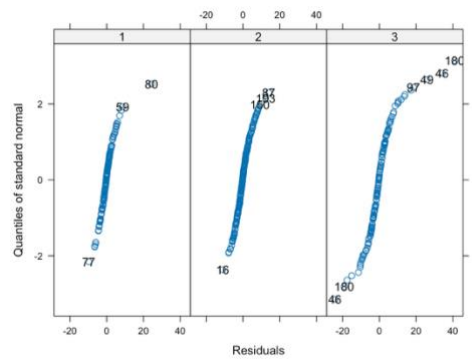
Table 6.8 shows the fitted values compared to standardized residuals. Similar to the book, we can see that the variance decreases as fitted values increase, however now the group 1 and group 2's variances are slightly more spread out than the book's graphic. This is due to them

allowing to have their own fitted variance, which is smaller than that of group 3, but independent, so the residuals scale better not to overcompensate for the extreme spread in group 3.

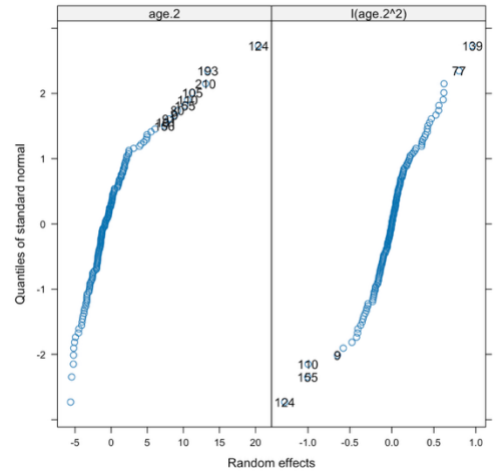
The residuals as grouped by age did not change significantly compared to the book's plot. While the added cubic term helped the mean, the presence of outliers still remains about the same, telling us that the cubic term helped to more closely fit the data closer to the mean of the VSAE score, but didn't affect the outlying data much.



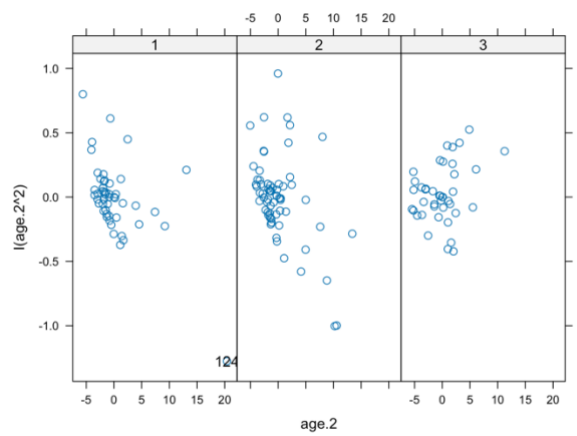
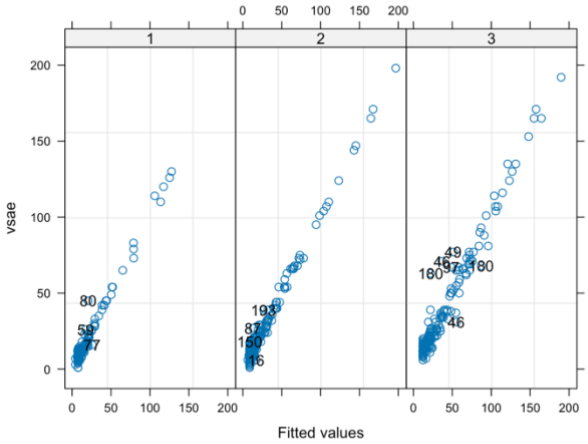
It's hard to get a sense of the spread in the Q-Q plots in the book since the graphics are very smushed, so I tried to replicate the image as close as I could. Overall the fit seems about the same. Most noticeably, there are now a few more listed outliers in the groups 1 and 2 categories (since their variance changed so we can now recalculate outliers), most noticeably 80 became more of an outlier than it originally was in the book's graph. More exploration is needed on this point. There are a few more outliers at the tails of group 2, but there are actually less outliers in group 3. Overall as group 3 gets further from the mean, the outliers still tail off much more.



The inclusion of the cubic fixed effect had little effect on the EBLUPs for the random effects, however further consideration might lead us to test whether a random variance term for the cubic term would be significant. Overall the cubic term had a much more drastic effect on the higher side of the age.2 random factor, and messed with both tails of the age.2^2 factor. Compared to the book, the EBLUPs for age.2 seem a bit worse to be considered normal, but the squared term's normality assumption improved slightly. Further testing would certainly be needed, especially for the assumption of normality of the newly fitted EBLUPs for the age.2 term



The pairs plot didn't change all that much, the most change was in the group 3 category. Groups 1 and 2 didn't change, but the spread of group 3 was reduced significantly. This is because the inclusion of the cubic term is able to capture more data than the quadratic term on the mean, leaving the quadratic EBLUPs to more accurately reflect the effect of the quadratic term on the mean. The random quadratic term appears to be fit a little better than without the cubic fixed effect term.



Finally, the Agreement of observed VSAE scores with conditional predicted scores by SICDEGP level. These graphs did not change at all, meaning our overall conditional predictions didn't really get better or worse. This makes sense since we didn't mess with the random structure. Hypothetically we would expect there to be slightly less outliers, but it seems to not help as much (conditionally on the child) to make a difference in this graph, however it significantly helped our unconditional predictions (adding a new child, for example).